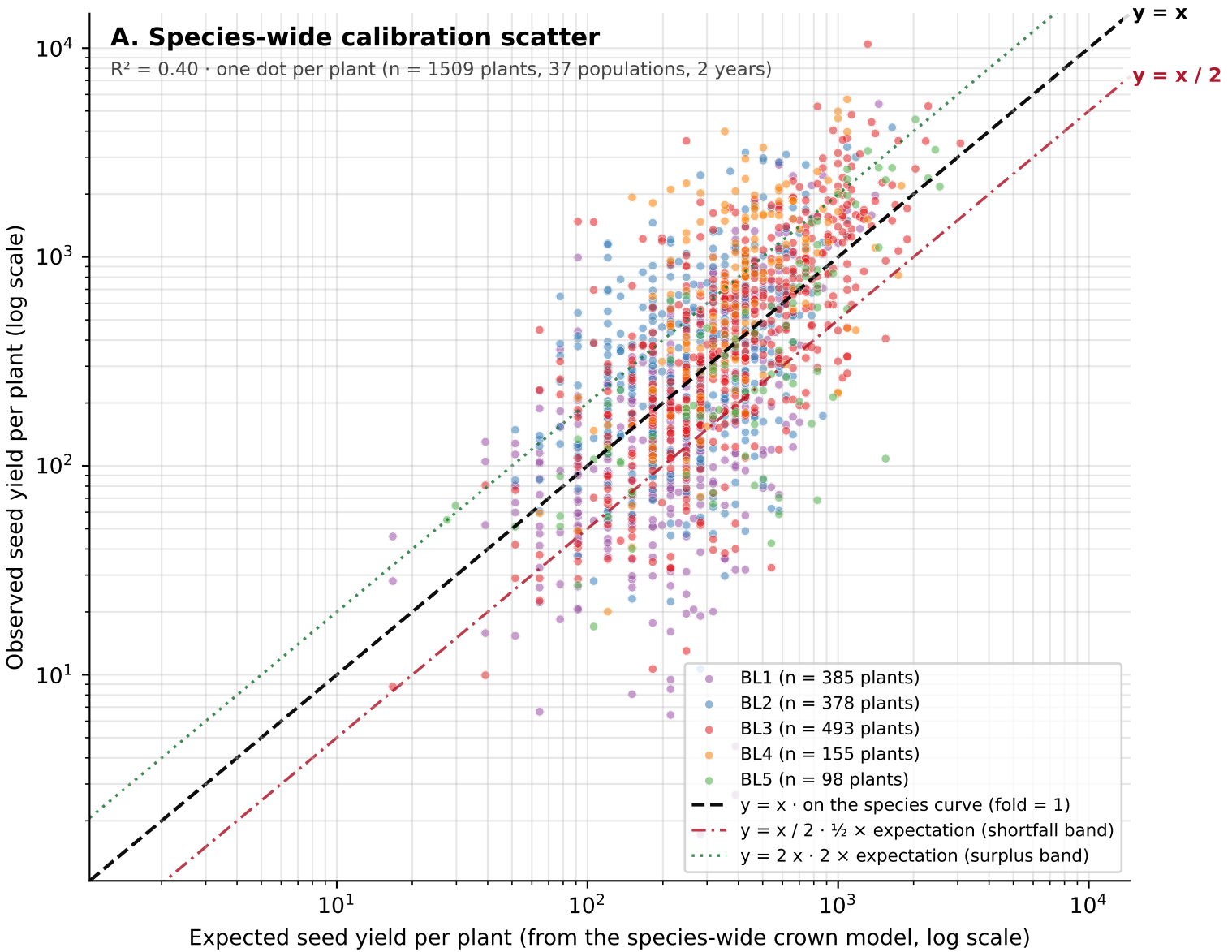
Species-wide plant-size → seed-yield expectation (the yardstick for the per-population seed-yield test)



B. Model build — how the species-wide expectation is set

STAGE 1 · predictor selection (10-fold cross-validation)

Lower CV-RMSE = better out-of-sample fit → wins Stage 2.

predictor	CV-RMSE	±sd	R^2_in	AIC
crown	0.409	0.026	0.395	1587.3 ★
crown+height	0.409	0.025	0.395	1589.2
area	0.418	0.025	0.366	1659.0
height	0.450	0.024	0.268	1875.8

→ Winner: crown (★)

STAGE 2 · refitted expectation model (all plants pooled)

Fitted value = expected $\log_{10}$(yield) for a plant of that size.
Residual = observed − expected.

Formula : $\log_{10}(\text{yield}) \sim \text{logc}$
Coefficients : Intercept = +1.223, logc = +1.227
R^2 (in sample): 0.395
Residual SD : 0.409 $\log_{10}$ (≈ × / ÷ 2.6)

Every population in the per-population test is measured against this expectation.